

LJUBLJANA BRNIK, Slovenia

WMO: 140140

Lat: 46.218N

Lon: 14.473E

Elev: 388

StdP: 96.75

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-9.9	-15.2	1.0	-8.8	-12.9	1.3	-6.5	7.2	2.7	5.9	1.0	0.8	290	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	31.0	20.9	29.3	20.3	27.8	19.8	21.9	28.7	21.1	27.6	20.2	26.4	2.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	15.1	25.2	18.8	14.3	24.2	18.0	13.6	23.1	65.8	28.9	62.7	27.8	59.7	26.6	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.0	4.2	-16.2	33.4	3.7	1.9	-18.9	34.7	-21.0	35.8	-23.0	36.9	-25.7	38.3		
(5)			-16.6	23.5	3.6	1.1	-19.1	24.3	-21.2	24.9	-23.2	25.5	-25.8	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-0.7	0.9	5.2	9.8	14.3	18.3	20.0	19.7	14.9	10.4	5.4	0.0		
	DBStd	8.18	4.18	4.03	3.75	3.31	3.07	3.23	2.85	2.92	3.12	3.68	4.31	4.31		
	HDD10.0	1284	332	254	153	43	3	0	0	0	2	41	147	310		
	HDD18.3	3258	591	487	408	256	128	40	16	19	109	246	389	568		
	CDD10.0	1246	0	0	3	37	137	249	311	300	149	53	7	0		
	CDD18.3	178	0	0	0	0	4	39	68	61	7	0	0	0		
	CDH23.3	2008	0	0	0	8	99	455	741	633	71	1	0	0		
	CDH26.7	584	0	0	0	0	13	124	236	204	6	0	0	0		
(14) Wind	WSAvg	1.4	1.1	1.3	1.6	1.7	1.7	1.6	1.5	1.4	1.3	1.2	1.2	1.0		
(15) Precipitation	PrecAvg	1514	82	88	101	111	116	142	133	135	157	164	169	117		
	PrecMax	2121	256	306	233	202	229	230	254	258	390	447	445	281		
	PrecMin	1178	4	4	13	8	39	49	49	28	40	5	10	0		
	PrecStd	221	53	70	62	47	46	41	50	63	82	97	95	68		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	15.4	19.9	24.5	28.2	31.8	33.2	33.8	27.6	22.5	16.9	12.0		
		MCWB	8.4	8.1	11.1	14.5	18.4	21.0	21.2	21.4	19.7	16.7	13.2	10.1		
	2%	DB	9.0	11.3	17.1	21.9	25.9	29.9	31.0	31.0	25.5	20.2	14.4	9.9		
		MCWB	6.8	6.8	10.0	13.0	17.1	20.4	21.0	20.9	18.7	15.8	12.2	8.2		
	5%	DB	7.2	9.1	14.8	19.9	24.0	28.0	29.2	29.0	23.3	18.2	13.0	7.9		
		MCWB	5.6	6.1	9.1	12.4	16.2	19.6	20.3	20.4	17.6	14.5	11.4	6.9		
	10%	DB	5.7	7.1	12.3	17.8	22.1	26.1	27.6	27.0	21.4	16.4	11.7	5.9		
		MCWB	4.5	4.8	7.8	11.5	15.3	18.7	19.9	19.7	16.5	13.6	10.3	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.2	12.1	15.6	19.5	22.5	23.1	23.0	20.6	17.5	13.9	10.6		
		MCDB	10.0	12.8	17.5	22.4	26.1	29.2	30.7	30.2	26.0	21.3	15.6	11.8		
	2%	WB	7.3	7.7	10.7	14.0	18.0	21.2	21.9	21.7	19.2	16.3	12.7	8.5		
		MCDB	8.5	10.3	15.8	20.0	24.5	27.9	28.9	28.5	24.4	19.5	14.2	9.5		
	5%	WB	6.0	6.5	9.6	12.9	16.8	20.1	21.0	20.9	18.2	15.2	11.7	6.9		
		MCDB	6.8	8.9	13.6	18.3	22.7	26.6	27.6	27.4	22.5	17.6	13.0	7.6		
	10%	WB	4.4	5.0	8.6	11.9	15.8	19.2	20.2	20.0	17.1	14.1	10.4	5.2		
		MCDB	5.2	6.9	11.8	16.7	20.9	25.2	26.3	26.1	20.6	16.2	11.5	5.7		
(35) Mean Daily Temperature Range	MDBR		7.5	9.6	11.6	12.0	12.4	12.1	12.7	12.4	11.0	10.0	7.1	6.6		
	5% DB	MCDBR	9.8	13.5	16.7	17.1	16.4	15.4	15.7	15.9	13.9	12.4	9.4	9.1		
		MCWBR	7.4	9.2	10.2	9.3	8.3	7.2	6.9	6.9	7.7	8.0	6.9	7.3		
	5% WB	MCDBR	8.3	12.1	14.1	14.8	14.8	14.5	14.5	14.6	12.6	10.9	7.9	7.8		
		MCWBR	6.7	8.6	9.0	8.3	7.8	7.1	6.8	6.9	7.2	7.2	5.9	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.346	0.393	0.443	0.444	0.452	0.449	0.443	0.419	0.393	0.353	0.317		
	taud		2.402	2.325	2.225	2.114	2.160	2.186	2.203	2.232	2.266	2.308	2.374	2.450		
	Ebn at Noon		774	818	824	813	826	820	817	805	789	750	718	736		
	Edh at Noon		72	94	120	147	145	143	138	130	115	96	74	63		
(44) All-Sky Solar Radiation	RadAvg		1.27	2.08	3.24	4.25	5.21	5.88	5.91	5.18	3.66	2.38	1.31	1.05		
	RadStd		0.18	0.34	0.44	0.50	0.51	0.39	0.39	0.56	0.48	0.28	0.20	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.55	N/A	N/A	+1.06	+0.54	+0.55	N/A	-160	+105	+66
(47) Regional (77 neighbors)		+0.48	N/A	N/A	+0.80	+0.45	+0.35	-80	-127	+95	+62

Nomenclature: See separate page